Vultr-Placeholder-Test123

A backend for AI agents that do real work:
a shell and a browser, sealed in one box.

Vultr hackathon, Blast Radius Zero track

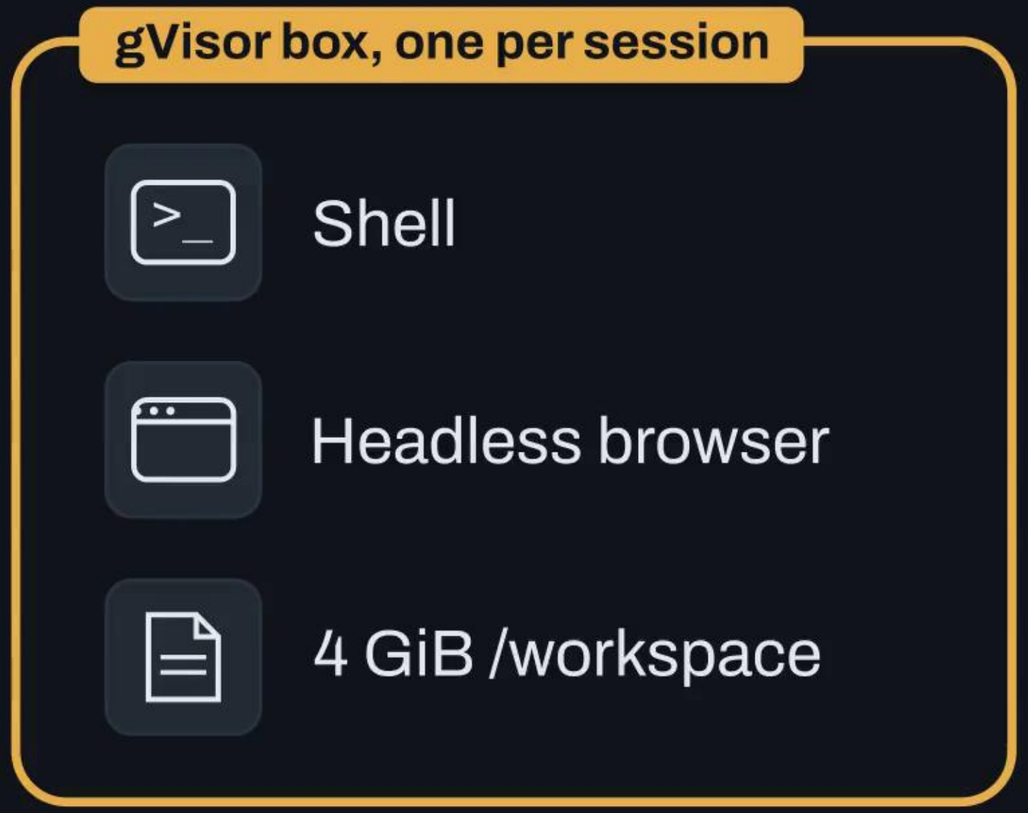


Useful agents need a shell and a browser



One prompt injection away from **all of it**.

Give every session its own sealed box



One sealed box per session

gVisor, uid 1000, read-only root, no capabilities, capped CPU, memory and disk

No credentials inside

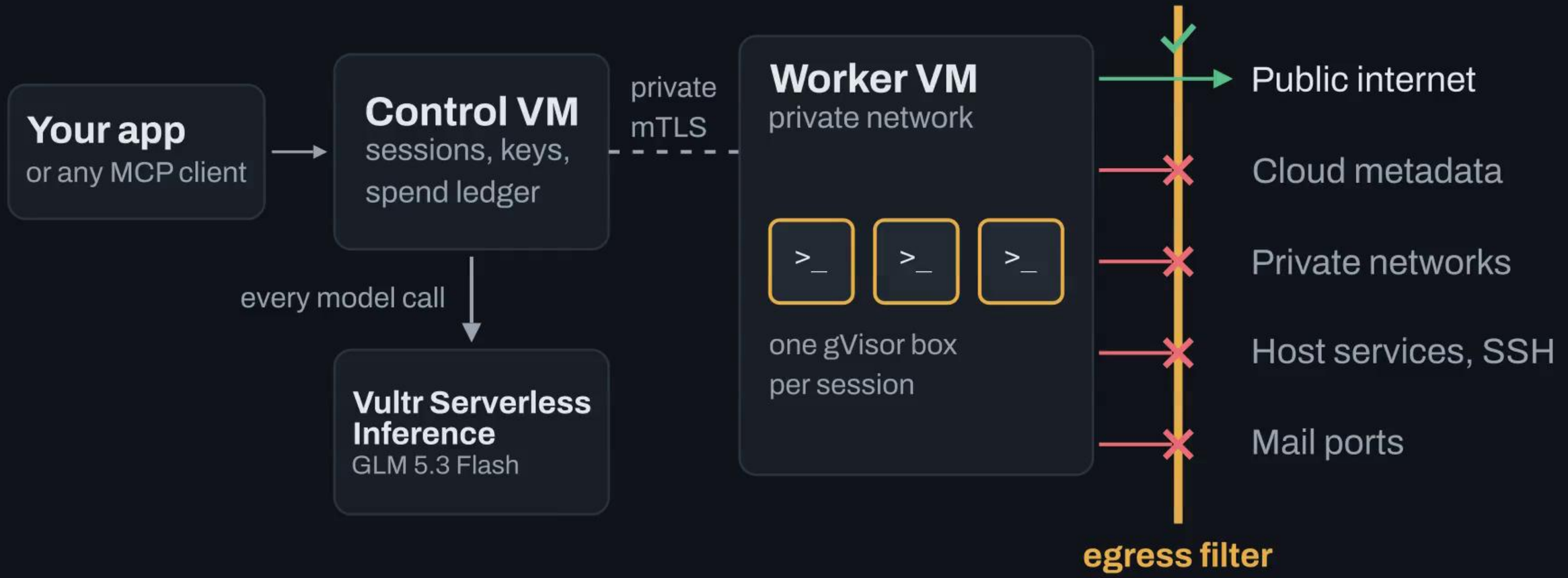
API keys, the worker token and the inference key stay in trusted code

Public internet only

Cloud metadata, private networks, the host and mail ports are blocked

How it fits together

Two VMs, one wall



What agents can do

Real tools, inside the box

in the box

Shell

Build, test, install, start servers

in the box

Browser

Clicks, types and takes screenshots

in the box

Files

Read, edit and search files in /workspace

in the box

Web research

Searches the web and reads pages as text

in the box

Convert

pandoc, LibreOffice, ImageMagick, ffmpeg

trusted code

Skills

Loads and saves SKILL.md playbooks

trusted code

MCP

Remote servers on the operator's list

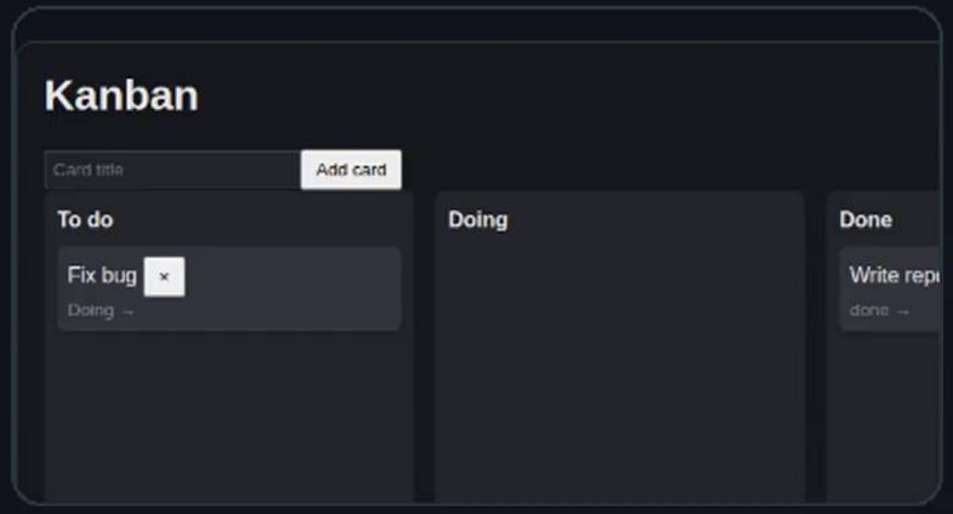
trusted code

Its own config

Can tune itself, not its own limits

Live demo, session 629d0b6f

Build a kanban app, then use it



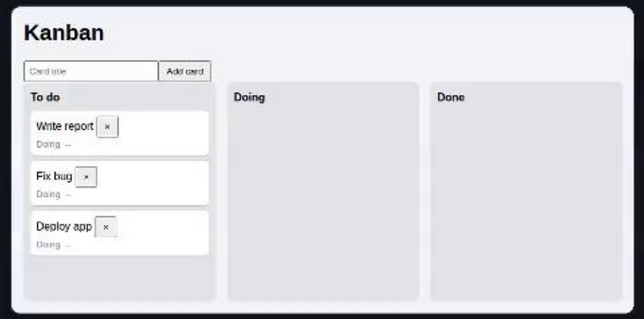
Its steps in the web chat

166 s
build and use

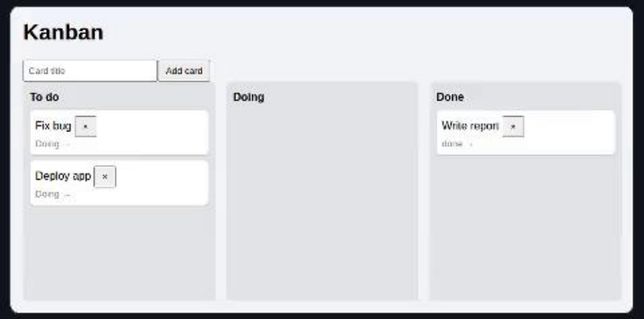
82 s
dark mode

16 s
delete

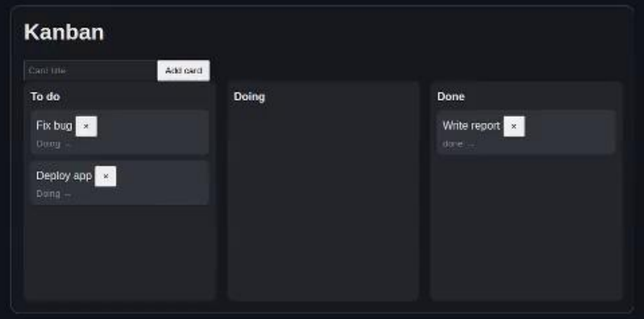
3 turns, 92 tool calls, $0.0244 model cost



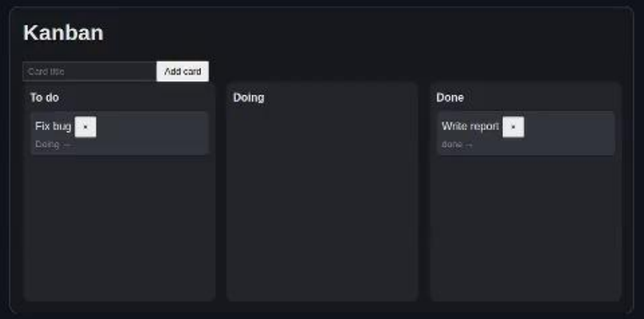
Three cards added



Moved to Done, kept after reload



Dark mode toggle, turn 2



A card deleted, turn 3

Live demo, session c8fcac86

Data, a chart, a memo, tidy files

3.9 s Totals per region from sales.csv
36 rows; North 2,125,200, best month 2025-12

4.5 s Bar chart, revenue.png
First try; the bars match the CSV totals

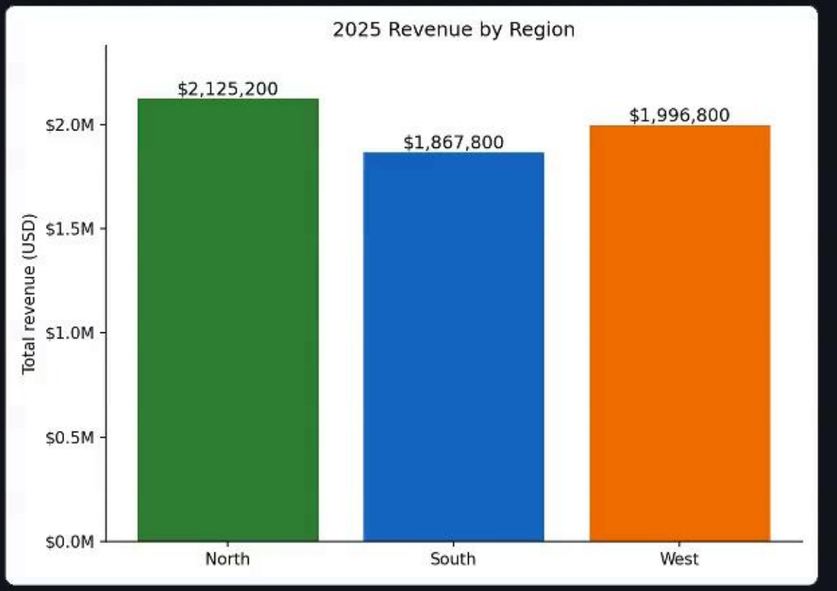
9.3 s Memo as .docx plus a PDF copy
Fell back to LibreOffice when pdflatex was missing

2.0 s Rename 5 messy file names
To lowercase-with-dashes

1.7 s PNG to an 800 px JPG
40,020 B down to 34,193 B

5 turns, 7 tool calls, $0.0034 model cost

The memo's revenue figures are right; its units and softest month were invented by the model, never computed.



revenue.png, drawn by the agent with matplotlib

- revenue.png: 40,020 bytes (~39 KB)
- revenue.jpg: 34,193 bytes (~33 KB)

revenue.jpg

Turn 5 in the web chat, both sizes reported

Live demo, session 1c047b6c

Research, build, test, report

Research
`web_search, web_fetch`

→

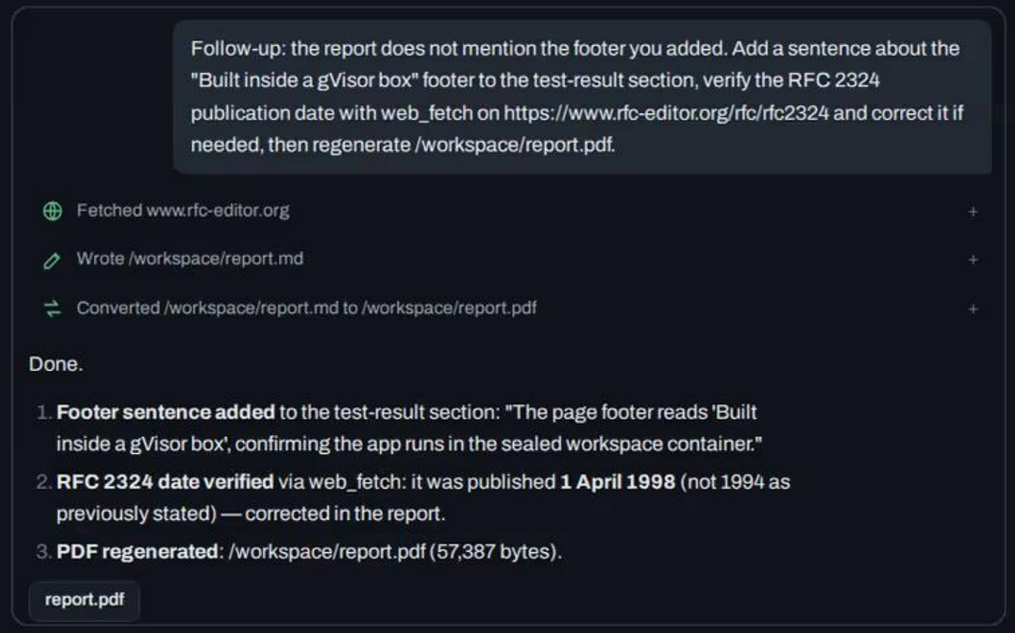
Build
`write_file, bash`

→

Browser test
`type, click, screenshot`

→

PDF report
`convert_file`



Follow-up turn in the web chat, with the PDF attached



The app, as the agent's browser saw it

35 s
first turn

17
tool calls, two turns

$0.0054
model cost

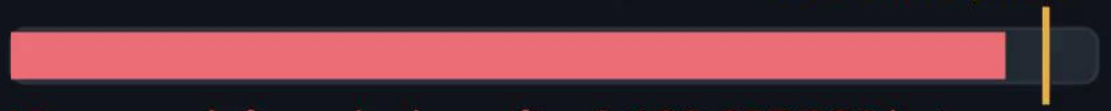
Containment, tested live

We tried to break out

Probe session 2401b687, real tool calls

- ✕ Cloud metadata, 169.254.169.254
  No route to host
- ✕ Worker listener, 10.42.0.4:8443
  Could not connect
- ✕ Host SSH, port 22
  Could not connect
- ✕ SMTP, ports 25 and 587
  Network is unreachable

dd 4300 MiB into /workspace 4096 MiB quota



No space left on device, after 4,126,887,936 bytes

9/9 box checks pass

- ✓ Metadata
- ✓ Private networks
- ✓ Host services
- ✓ SMTP
- ✓ Public HTTPS
- ✓ No credentials
- ✓ Read-only root
- ✓ Disk quota
- ✓ Volume destroyed

10/10 job checks pass

Integrate it

One HTTP call to start an agent

REST: create a session with its first message

```
curl $HOST/v1/sessions \
  -H "authorization: Bearer $AR_KEY" \
  -H 'content-type: application/json' \
  -d '{"message":"Build a kanban board and test it"}'
```

MCP: add it to Claude Code, Cursor or any HTTP MCP client

```
claude mcp add --transport http agent \
  https://<host>/mcp \
  --header "Authorization: Bearer ar_..."
```

REST API at /v1
Sessions, messages, files

Live event stream
Server-sent events, resumable

Signed webhooks
HMAC-SHA256 on every turn end

MCP server at /mcp
12 tools, the same API keys

Built on Vultr

Every agent call runs on Vultr

AI

Vultr Serverless Inference

Every agent model call, GLM 5.3 Flash

VM

Two Vultr VMs in Chicago

Control 2 vCPU, 4 GB. Private worker 4 vCPU, 8 GB. About $0.082 per hour

Model cost of real runs on the public instance

$0.0244
Build and use a kanban app
92 tool calls, 3 turns

$0.0054
Research, build, test, PDF
17 tool calls, 2 turns

$0.0058
One-shot job 7d466fa9
5 model calls, 58 s

Vultr-Placeholder-Test123

Agents with a shell and a browser.
A blast radius of one box.

Try it live

`https://144-202-59-108.sslip.io`

